Highlights

Optimizing warnings from hydrologic ensemble prediction systems for improved decision making

Jesús Casado-Rodríguez, Peter Salamon, Stefania Grimaldi, Corentin Carton De Wiart, Christel Prudhomme, Calum Baugh, Ervin Zsoter, Ilias Pechlivanidis, Nina Bosshard, Michaela Mikulickova

- Probabilistic NWP models outperform deterministic counterparts in terms of flood warning skill.
- Using persistence (consecutive forecasts) as a warning criteria is not effective in probabilistic systems.
- Building grand ensembles adds value to the skill of the best performing NWP but this added value decreases with lead time.

Optimizing warnings from hydrologic ensemble prediction systems for improved decision making

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Abstract

Early Warning Systems are critical in mitigating the impacts of floods, the most frequent and economically damaging natural hazard worldwide. These systems rely on hydrological simulations driven by meteorological forecasts, both of which are inherently uncertain. Ensemble Prediction Systems (EPS) were developed decades ago to account for these uncertainties by generating multiple future scenarios that portray the forecast uncertainty. The advantages of probabilistic modelling have been proven over the last decades both in the fields of meteorology and hydrology. In order to issue a flood warning, the probabilities provided by EPS need to be converted into dichotomous values of "warning" or "not warning". This conversion is often done by application of thresholds, whose definition is pivotal for the final skill of the system. While meteorological and hydrological models are under continuous development —higher spatial and temporal resolution, better process representation, improved calibration—, the methodologies for converting discharge forecasts into flood warnings lack continuous evaluation. In this paper, we use discharge simulations from the European Flood Awareness System (EFAS) to assess the skill of the four NWP models of different nature —probabilistic and deterministic—within the system, and search for methods to combine all the NWP into a grand ensemble that enhances overall skill. Additionally, we evaluate the effects of the current notification criteria —probability threshold and persistence— and determine their optimal values. Our results demonstrate that probabilistic NWP models outperform deterministic ones, which demonstrate skill only at short lead times. Additionally, our findings suggest that the persistence criterion is beneficial for removing inconsistencies between consecutive forecasts in deterministic NWP, but should be omitted from probabilistic systems. By applying optimal notification criteria to a single probabilistic NWP (ECMWF-ENS), we observe an improvement of up to 6.4% in skill compared with the current EFAS procedure, which employs four NWP models. When compared with the optimized ECMWF-ENS warnings, grand ensembles only yield significant improvements in skill (up to 4.2%) within the first two days lead time. Two methods to build the grand ensemble stand out —equiprobability among all model runs or skill weighted—; we discuss advantages and drawbacks of these two approaches. Overall, our study underscores the importance of selecting appropriate flood notification criteria and limits the value added to the grand ensemble by deterministic NWPs to the shortest lead time range.

Keywords: EFAS, early warning system, flood warnings, probabilistic forecasting

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1. Introduction

Floods are the natural disaster that causes the biggest economical losses in Europe WMO (2021). In the European Union (EU), flood damages amounted to a total cost of approximately EUR 280 billion in the period 1980-2022, representing 43% of all meteorological hazards EEA (2023). The historical data show an increasing tendency in flood damages over Europe EEA (2023); Paprotny et al. (2018), and climate projections indicate an exacerbation of this trend with increasing frequency and severity of extreme events Alfieri et al. (2017); Martina Angela Caretta et al. (2023). In this scenario, Flood Early Warning Systems (FEWS) emerge as one of the multiple adaptation measures to be taken. The major flood event in the Elbe River in 2002 triggered the development of the European Flood Awareness system (EFAS), a continental scale FEWS that has been operational since 2012 as part of the Copernicus Emergency Management Service and whose monetary value has been proven Pappenberger et al. (2015). The magnitude of the flood that hit Germany and Belgium in July 2021, the most costly in the last 40 years, EEA (2023) was a reminder of the importance of FEWS and their need for continuous improvement. On a global scale, the United Nations launched the initiative Early Warnings for All to ensure that the entire World population is warned about weather, water and climate hazards by 2027 Egerton et al. (2023).

FEWS rely on meteorological forecasts generated by Numerical Weather Prediction (NWP) models. Over the past thirty years, Ensemble Prediction Systems (EPS) have been developed within the realm of NWP models to address uncertainties in the initial conditions. EPS provide equiprobable forecasts at any given time and location, enabling the quantification and dissemination of probabilistic forecasts Cloke and Pappenberger (2009). In comparison to deterministic models, EPS have demonstrated enhanced consistency across consecutive forecasts Buizza (2008) and improved precipitation forecasts at longer lead times Molteni et al. (1996).

Approximately two decades ago, the success of EPS in meteorology led to the adoption of this paradigm in flood forecasting Diomede et al. (2008); Bartholmes et al. (2009); Thielen et al. (2009); Yang et al. (2020). Hydrological Ensemble Predictions Systems (HEPS) propagate the uncertainties in the meteorological forecasts by forcing hydrological models with the members of a meteorological ensemble. Although this approach is the most common, it only considers the uncertainty in the initial meteorological condition, but does not include other sources of uncertainty, such as model representation or model parameterization Cloke and Pappenberger (2009).

The decision to issue weather-related warnings, such as floods, is delicate as it might result in significant costs, either due to unpreparedness if an event is missed or the mobilization of resources in false alarms. Weather forecasting systems often issue a higher number of false alarms than missed events Bouttier and Marchal (2023), despite the potential risk of eroding trust in the system, known as the 'cry wolf' effect LeClerc and Joslyn (2015). This latter research showed that an increase in false alarms diminishes the credibility of the system, leading end-users to disregard precautionary measures. However, they also demonstrated that incorporating uncertainty estimates in warning messages can assist users in making informed decisions. Therefore, HEPS have the potential to enhance both the skill and trustworthiness of flood warning system, if particular attention is given to the balance between false alarms and missed events.

EFAS is the hydrological forecast component of the Copernicus Emergency Management Services (CEMS) of the European Commission Thielen et al. (2009). EFAS is an operational system that monitors and forecasts floods in an extensive European domain, encompassing not only European Union member states. Its primary objective is to enable proactive measures in anticipation of significant flood events. EFAS functions as a complementary system to national or regional counterparts, with a specific focus on trans-boundary rivers and medium-range forecasts.

With the release of version 4.0, EFAS has been generating forecasts twice a day, with a 6-hour resolution and lead times extending up to 10 days. It couples four weather forecasts of diverse nature —deterministic and probabilistic— with the distributed, physically-based hydrological model LISFLOOD-OS Burek et al. (2013). The simulation produces a set of water fluxes and state variables, from which river discharge is utilized to issue flood warnings. EFAS employs a threshold exceedance approach for issuing warnings Bartholmes et al. (2009), where continuous discharge time series are converted into binary series of exceedance or not-exceedance over a discharge threshold (the 5-year return period). In their work, Bartholmes et al. Bartholmes et al. (2009) developed a notification criteria based on two types of persistence: persistence in a particular forecast is determined by the number of NWP members predicting the exceedance, while persistence over several forecast establishes that a flood signal is reliable when consecutive forecasts predict the event. In this study, we will use the term "probability of exceedance" to refer to the first type of persistence, as the number of NWP members exceeding the discharge threshold can be translated into a probability, assuming equiprobability among them). We will keep the term "persistence" to denote only the consistency among consecutive forecasts.

The criteria applied in EFAS to issue flood warnings, so-called notifications, have evolved over the past years and currently a 30% probability threshold and a persistence of 3 consecutive forecasts are used. These criteria are independently applied to the two deterministic and the two probabilistic NWP models, issuing a notification only if at least one of the models of each type predicts the flood. Additionally, EFAS notifications are exclusively dispatched to locations with a minimum catchment area of 2000 km^2 and for lead times longer than 48 hours. The first limit is based on the inferior hydrological performance in smaller catchments, while the second limit is a consequence of EFAS serving as a supplementary system to national or regional services. Previous studies have examined the skill of EFAS notifications Bartholmes et al. (2009) and the response of end users Demeritt et al. (2013). However, the continuous advances in NWP performance Bauer et al. (2015), as well as in the resolution and representation of hydrological processes in EFAS Joint Research Centre - European Commission (2020, 2023), requires a new evaluation of the notification criteria to harness the potential of the system.

The objective of this analysis is to assess the skill of a continental flood forecasting system like EFAS in its current state, and explore new notification criteria that can maximize its warning skill. In particular, the first goal is to determine the value of combining several NWP models of different nature in a single system, specifically seeking the appropriate combination method to build a grand ensemble. The second goal is to optimize the two notification criteria —probability threshold and persistence—, analysing the interplay between them and reconsidering whether the persistence criterion is meaningful in a purely probabilistic approach. The third and final goal is to analyse the evolution of the notification skill with catchment area to discern if the newer, higher resolution models are skillful in smaller catchments.

To address our research questions, we conducted an analysis of EFAS4 discharge simulations for the period 2020-2023. Section 2 enumerates the datasets used in the analysis. Section 3 explains the two experiments that analyse the skill of individual NWP models and different combinations of those models, the criteria optimization process, and the skill metrics we targeted. Section 4 follows a similar structure, presenting first the skill of individual NWP, followed by that of the combination methods, and ends with an exploration of the relationship between notification skill and catchment area. Section 5 discusses the importance of the spatio-temporal framework and target metric in skill analysis like the one here presented, the trade-offs between the notification criteria, the benefits of combining NWP models, and the challenges in selecting a combination method.

2. Data

The data used in this analysis is taken from the operational setup of the EFAS in its version 4 Joint Research Centre - European Commission (2020). EFAS generates a hydrological forecast twice a day (00 and 12 UTC) forced by the meteorological forecast of four different NWP models outlined in Table 1. EFAS integrates forecasts from probabilistic models (COS and ENS) and deterministic models (DWD and HRES) to issue flood warnings.

The study period spans from the release of EFAS4 in October 2020 until June 2023. For this period we use the EFAS forecasts as the discharge prediction and the EFAS reanalysis as a proxy of discharge observations. Using

Table 1: Characteristics of the numerical weather prediction models used in EFAS4.

Model	Provider	Acronym	Max. lead time	No. ensembles	Spatial resolution
COSMO-LEPS		COS	5.5 days	20	$\sim$ 7 km
ICON-EU/ICON	DWD	DWD	7 days	1	$\sim$ 6.5-13 km
HRES	ECMWF	HRES	10 days	1	$\sim$ 9 km
ENS	ECMWF	ENS	10 days	51	$\sim$ 18 km

reanalysis as ground truth avoids the limitation of observational data (e.g. gaps in the observed time series, difference in recorded period...), and reduces model representation and calibration errors from the analysis. The previous data sets were converted from a continuous variable —discharge— into binary events of exceedance or not exceedance over a threshold. Following the current EFAS setup, the flood threshold is the discharge associated with the 5 year return period, which was computed by fitting a Gumbel distribution to the annual maxima from the discharge simulated in the EFAS4 long-term run (1991-2023) .

In contrast to the original EFAS skill assessment Bartholmes et al. (2009), the current analysis focused solely on real gauging stations rather than all river cells in the model, which would have cause an autocorrelation issue. A selection was made from the approximately 4000 gauging stations in the EFAS database based on two conditions: the contributing area must exceed 500 km^2, and the hydrological performance, measured in modified KGE (Kling-Gupta efficiency) Kling et al. (2012); Gupta et al. (2009); Knoben et al. (2019), must not fall below 0.50. The lower bound on catchment was set due to the relatively coarse spatial resolution of NWP and EFAS temporal resolution (6 h), which limits hydrological performance in small catchments. The condition based on hydrological performance ensures the validity of using model simulations as proxies for observations is valid. Ultimately, 1979 gauging stations were included in the analysis, accounting for a total of 1683 events (exceedances over the 5 year threshold) during the study period. As explained in section 3.4, the optimization of the notification criteria was conducted on a subset of 1239 stations with a contributing area larger than 2000 km^2 (874 events), while the entire set was used to analyze the influence of catchment area on warning skill.

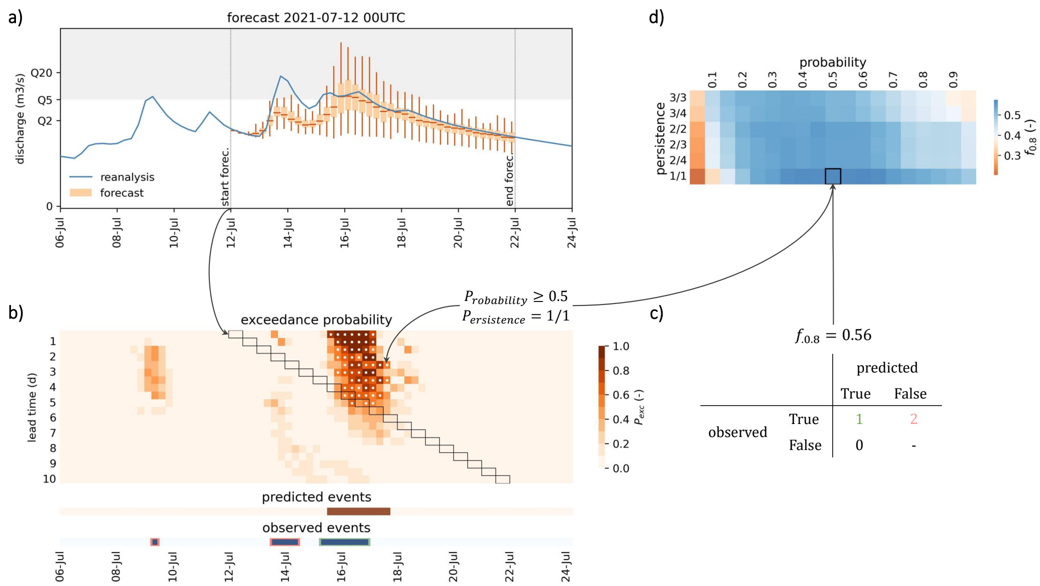

Figure 1: Overview of the skill assessment framework. a) Hydrographs comparing observed and forecasted discharge with the discharge associated to the 5-year return period. b) Matrix of total exceedance probability (white dots identify pairs of date and lead time that meet the notification criteria, whereas black rectangles delineate the location of the forecast hydrograph from panel (a) in the probability matrix) and identification of predicted events for specific values of probability threshold and persistence. c) Confusion matrix for the selected station, constructed based on the previous notification criteria . d) Evaluation of the f-score across all combinations of probability threshold and persistence; the black square highlights the skill score corresponding to the notification criteria in panel (b).

3. Methods

The skill assessment comprises four steps, as depicted schematically in Figure 1: (a) transformation of both observed and forecasted discharge time series into corresponding time series of the probability of exceeding the 5-year return period; (b) construction of an exceedance probability matrix that consolidates the overlapping forecasts, followed by identification of predicted events through the application of specific notification criteria on that matrix; (c) calculation of the contingency table —hits, misses and false alarms—, (d) evaluation of the skill of the system across all possible combinations of the notification criteria.

3.1. *Exceedance over threshold*

The notification process starts by converting the continuous time series of the target variable into time series of exceedance or non-exceedance over a predefined threshold. Within EFAS, flood notifications are based on discharge, and the critical threshold is the discharge associated with a 5-year return period (Q_5) generated from a 30-year historical simulation. While alternative alert systems might rely on disparate variables, such as precipitation Bouttier and Marchal (2023) or river stage Nevo et al. (2022), and employ different thresholds for defining events, the underlying methodology remains similar.

The application of the discharge threshold yields distinct outcomes when comparing a single time series—like observations or deterministic forecasts—with an ensemble forecast. In the former case, the time series is converted into a binary sequence, where 0 represents non-exceedance and 1 denotes exceedance. Conversely, ensemble forecasts lead to a probability time series, with values fluctuating between 0 and 1. Since forecasts —both deterministic and probabilistic— are generated more frequently (every 12 hours) than the forecast horizon, they overlap. To address this, the ensemble forecasts are restructured into an exceedance probability matrix that connects each time step to increasing lead times, as illustrated in panel (b) of Figure 1. Such a configuration simplifies both the analysis and the subsequent computations. Within this matrix, an individual forecast is represented by a diagonal (black rectangles). The uppermost rows represent shorter lead times, where the probability of accurate prediction generally increases. For instance, in the example in Figure 1, the flood event occurring around July 16 was predicted with a 50% probability five days in advance, increasing close to certainty within two days of the event.

3.1.1. *Combination of NWP*

One of the particularities of EFAS is the simultaneous use of four NWP, both deterministic and probabilistic. The purpose of using multiple NWP is to leverage the strengths of each model. Thus, a primary aim of this study is to develop an effective method for integrating deterministic and probabilistic forecasts. In practical terms, EFAS forecasts produce four distinct probability matrices, one for each NWP, in contrast to the single matrix depicted in panel (b) of Figure 1. The challenge lies in devising a methodology to combine these four matrices into a total exceedance probability matrix, thereby enhancing the system's predictive accuracy compared to the use of a single NWP. We explored three different approaches for calculating the total exceedance probability and compared them with EFAS current approach, which does not incorporate a total probability framework.

In the current EFAS procedure, a notification is issued if at least one deterministic and one probabilistic NWP predict the event. This approach will be referred to as *1 deterministic + 1 probabilistic* (1D+1P) from here on. As each model is evaluated independently, the current procedure does not calculate a total probability matrix. The most straightforward method to compute the total probability is the *model mean* (MM); it is a simple mean over the four exceedance probability matrices, meaning all models are given equal weight. In other words, the single member of a deterministic model is considered significantly more important than any members of a probabilistic model. An alternative approach would be to assign the same weight to each member, regardless of whether it belongs to a deterministic or a probabilistic model. This approach will be referred to as *member weighted* (MW). In this approach, each NWP receives a weight relative to the number of members it contains, meaning the probabilistic models have greater influence than the deterministic ones. However, neither of these approaches take into account the performance of the NWP. To address this limitation, the *Brier weighted* approach (BW) assigns a weight to each model based on its probabilistic skill. The Brier score Brier (1950) —a probabilistic error metric that ranges from 0 to infinite, with 0 being the optimal value— is used as a measure of probabilistic skill. The Brier score is calculated using the equation:

$$\mathrm{BS} = \frac{1}{T} \sum_{t=1}^{T} \left(P_{obs,t} - P_{pred,t}\right)^2 \tag{1}$$

where BS is the Brier score of a single station, T is the number of time steps, $P_{obs,t}$ is the observed probability of exceedance, and $P_{pred,t}$ is the predicted probability of exceedance at a specific time step. The results of the BW approach are highly sensitive to how Brier scores are converted into weights. The conversion function must transform the values from a metric whose optimal value is 0 to a weight with an optimal value of 1. Additionally, it should be exponential to amplify the differences among NWP, considering the extremely low BS values that are characteristic in rare events where both the observed and predicted probabilities are 0 most of the time. Inverse distance weighing (eq. 2) fulfils both conditions. Values of the exponent p from 1 to

Table 2: Contingency table in a binary classification. E_{obs} is the sum of observed events and E_{pred} the sum of predicted events.

		Observed		
		True	False	
Forecasted	True	hit	false alarm	E_{pred}
	False	miss	true negative	
		E_{obs}		

9 were tested, and 7 was found to be the optimal value.

$$w = \frac{BS^{-p}}{\sum BS{-p}} \tag{2}$$

The weights in the BW method are specific to each NWP and lead time, whereas in the MM and MW methods they remain stable over lead time.

3.2. Contingency table

The skill assessment of a warning system involves a binary classification task that compares two time series: the predicted and observed events. Typically, the contingency table (Table 2) is used to summarize performance in this type of classification problem. Given that floods are rare events, the classification task is highly imbalanced, resulting in the number of true negatives being orders of magnitude larger than any of the other terms in the contingency table. Consequently, we must omit the true negatives from both the computation of the contingency table and the selection of the target skill metric not to overestimate skill (Section 3.3).

The derivation of the binary time series of observed events is straight forward by application of the discharge threshold (Q_5) over the reanalysis time series. Conversely, deriving the binary time series of predicted events involves applying two notification criteria to the matrix of total probability of exceedance (or the individual matrices of exceedance probability for each NWP in the 1D+1P approach). The probability threshold sets the minimum value of the exceedance probability at which a high risk of flooding is considered and a notification must be issued. In the current EFAS setup, the threshold is 30% and applies only to the probabilistic forecasts. Additionally, the persistence criterion was introduced to remove false alarms caused by the erratic behaviour of some NWP Bartholmes et al. (2009). This criterion aims to replicate the behaviour of a forecaster, who would wait to send the warning until consecutive forecasts predict the event. In the current procedure, notifications are sent if 3 consecutive forecasts exceed the probability threshold (hereafter referred to as 3/3 persistence). Panel (b) in Figure 1 illustrates the application of these two notification criteria on the matrix of exceedance probability. For every time step (columns), the white dots identify the lead times that meet the criteria. Eventually, a flood event is predicted at a particular time step if any of the lead times meet the criteria.

The terms of the contingency table result from comparing the predicted and observed events. We consider a hit any observed event in which at least one time step was correctly predicted. The number of misses and false alarms are calculated, respectively, as the difference between the observed (E_{obs}) or predicted (E_{pred}) events and the hits (eq. 3).

$$\text{misses} = E_{obs} - \text{hits} \tag{3}$$

$$\text{false alarms} = E_{pred} - \text{hits} \tag{4}$$

The process is repeated for all possible combinations of the two criteria and across all stations. We tested probability threshold values ranging from 5% to 95% and six persistence values: 1/1 (no persistence), 2/4, 2/3, 2/2, 3/4 and 3/3 (the current persistence criterion). The outputs of this process are matrices of hits, misses and false alarms for each station an combination of the probability threshold and persistence that will be used in the subsequent section to compute skill metrics.

3.3. Skill metrics

The fact that the classification task we face is highly imbalanced is fundamental in the selection of the skill metric from the myriad of options. For instance, the odds ratio and the Hanssen-Kuipers score (HK) Hanssen and Kuipers (1965) were discarded because both include the true negatives, which is a term to be excluded in an imbalanced classification. Instead, we have selected three skill metrics specific for imbalanced classification, i.e., that exclude the true negatives.

Recall is the proportion of observed events that are correctly predicted (eq. 5); in some contexts it is also known as hit rate, probability of detection or sensitivity. Bartholmes Bartholmes et al. (2009) proved that recall is equal to HK for highly imbalanced classifications, such as floods. Precision, a.k.a. positive predicted value or frequency of hits, is the proportion of the predicted events that are correct (eq. 6). There is a well known trade-off between recall and precision. In the specific case of flood notifications, very strict criteria would cause very few notifications with high certainty, which means that they will be mostly correct (high precision), but a lot of events would be missed (poor recall). On the other hand, very relaxed criteria would cause a large amount of uncertain notifications, which means a small number of missed events (high recall) but a large number of false alarms (poor precision). As a compromise, the f_β score is a metric that balances recall and precision (eq. 7). In its most common version ($\beta = 1$) it is the harmonic mean of recall and precision, granting equal importance to both terms. However, higher importance can be given to precision ($\beta < 1$), therefore limiting the amount of false alarms, or to recall ($\beta > 1$), reducing the amount of misses. In our study, after suggestions from forecasters analyzing EFAS on a daily basis, we selected $f_{0.8}$ as the target skill metric. This metric balances precision and recall giving a slightly higher value to the former. The idea behind is to limit the amount of false alarms, which would jeopardize the trust of the recipients of the warnings. Both recall, precision and the f_β score range from 0 to 1, being 1 their optimal value.

As a final validation metric, we used bias, which represents the proportion between the number of predicted and observed events and is calculated as the quotient between recall and precision (eq. 8). Its optimal value is 1, which means that the number of notifications issued is equal to the number of observed events, and it ranges from 0 to infinity. All four metrics presented here can be plotted in the Roebber diagram Roebber (2009). We modified the original Roebber diagrams and replaced the critical success index by our target metric ($f_{0.8}$).

$$\text{recall} = \frac{\text{hits}}{E_{obs}} = \frac{\text{hits}}{\text{hits} + \text{misses}} \tag{5}$$

$$\text{precision} = \frac{\text{hits}}{E_{pred}} = \frac{\text{hits}}{\text{hits} + \text{false alarms}} \tag{6}$$

$$f_\beta = \left(1 + \beta^2\right) \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}} \tag{7}$$

$$\text{bias} = \frac{E_{pred}}{E_{obs}} = \frac{\text{hits} + \text{false alarms}}{\text{hits} + \text{misses}} = \frac{\text{recall}}{\text{precision}} \tag{8}$$

3.4. Optimal notification criteria

The current EFAS notification criteria involve 5 variables: minimum catchment area, minimum lead time, combination of NWP, probability threshold and persistence. We analysed and optimized all these variables, initially isolating each one to understand its individual effects.

We have conducted two main experiments to explore the benefits of combining NWP. The first experiment analyzes each NWP individually with two objectives: identifying the strengths and weaknesses of each NWP, and understanding how the notification criteria affect differently probabilistic and deterministic NWPs. The second experiment compares the four combinations of NWP explained in section 3.1.1. The aim is to identify the most appropriate combination method and compare it against the most skillful NWP to assess whether the combination of NWP adds value to the system. The setup of these two experiments is similar, apart from the difference in the input data, and is explained in the following paragraphs.

For the majority of the analyses, we used only stations that met the current minimum catchment area of 2000 km^2. With this subset of stations, we explored the evolution of skill with lead time, probability and persistence. The 20 lead time values —2 forecasts per day and an horizon of 10 days— were grouped in two ways. First, we grouped them daily to analyze the evolution of skill with lead time. Second, to explore the impacts of the probability threshold and persistence, we defined 4 lead time ranges: a) shorter than 2 days, a range at which EFAS formal notifications cannot be issued, b) 2-5.5 days, when all NWP are available, c) 5.5-7 days, when 3 NWP are available as the maximum lead time of COS is exceeded, d) 7-10 days when only 2 NWP are available as the maximum lead time of DWD is exceeded.

Following the exploration, we proceeded to optimize the probability threshold and the persistence criteria. Specifically, we sought the combination of probability and persistence that produces the maximum $f_{0.8}$ at each of the 4 lead time ranges previously specified. The optimization was repeated for every NWP individually and for every combination of the NWP. To ensure a robust selection of criteria, we applied a 10-fold cross validation: 20% of the stations were left aside as a test set, while the remaining 80% were subdivided in 10 folds. We computed the skill for every combination of 9 folds and then averaged over folds. The optimal notification criteria were derived from this average-over-fold skill matrix, as shown in panel (d) of Figure 1). Subsequently, the selected criteria were evaluated on the test set to prevent overfitting.

Lastly, we explored the impact of catchment area on warning skill using the entire set of stations. We calculated the notification skill for increasing values of the minimum catchment area, ranging from 500 km^2 to 300,000 km^2, using the criteria previously optimized ~~for the minimum catchment area of 2000 km^2~~. ~~Subsequently, we tested whether the skill of the system, particularly that of small catchments, significantly improves with a more complex notification criteria in which the probability threshold varies with catchment area. We fixed the persistence criterion to that obtained in the initial optimization and only tuned the probability threshold for increasing values of the catchment area threshold. We compared the skill of the fixed probability threshold with the skill of the area-specific probability threshold to evaluate whether the new, more complex criteria pays off in terms of performance.~~

4. Results

4.1. Analysis of "observed" events

Figure 2 depicts the spatial distribution of the 1239 stations used in the optimization of the notification criteria, i.e., those with a minimum contributing area of 2000 km^2 and a KGE not lower than 0.50. The colours represent the amount of "observed" flood events, while the histogram at the bottom shows the frequency of events across the stations.

In total, the optimization of the notification criteria will be based on 1239 stations and 874 "observed events". A higher concentration of points with events is observed in Central Europe, the British Isles, and Mediterranean catchments. Significant flood events in the Rhine, Meuse, and Ebro are visually evident on the map. Throughout the study period, 61% of the stations did not exceed the 5 year return period, which means that hits and misses can not exist, and the only term in the contingency table will be false alarms. Consequently, the f-score for these stations can be either 0 or null. Additionally, 29% of stations registered only one "observed" event. These nuances bear implications for the overall skill of the system, as we will explore in subsequent discussions.

The imposition of a minimum contributing area of 2000 km^2 places a constraint on the number of stations available for optimization, consequently affecting the count of observed events. In the final phase of our analysis, we assessed the implications of this criterion by expanding our consideration to stations with a minimum area of 500 km^2. Figure 3 represents the relationship between an increasing catchment area threshold and the corresponding

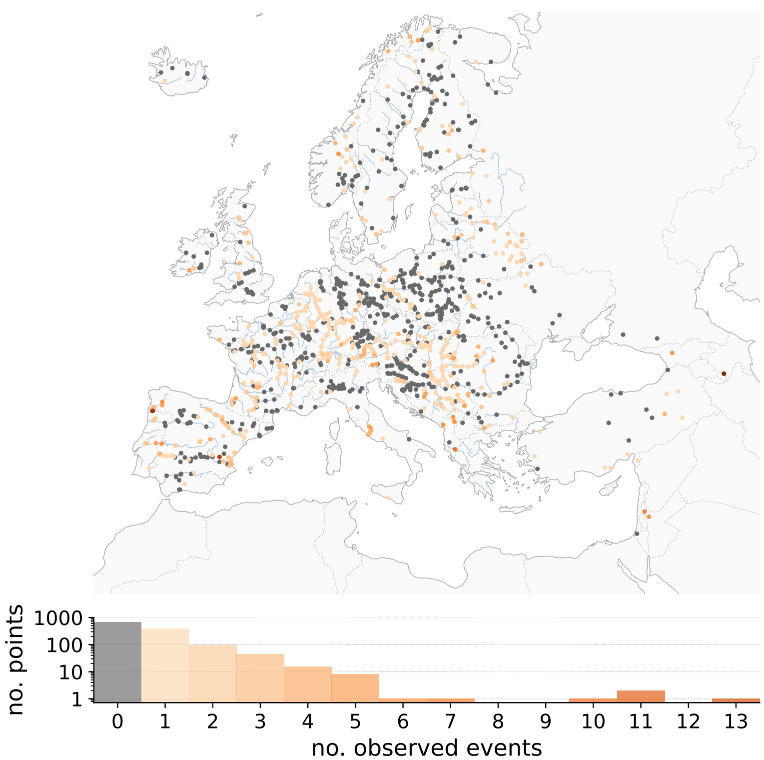

Figure 2: Geographical distribution of the stations used for optimizing the notification criteria, and number of "observed" flood events during the study period.

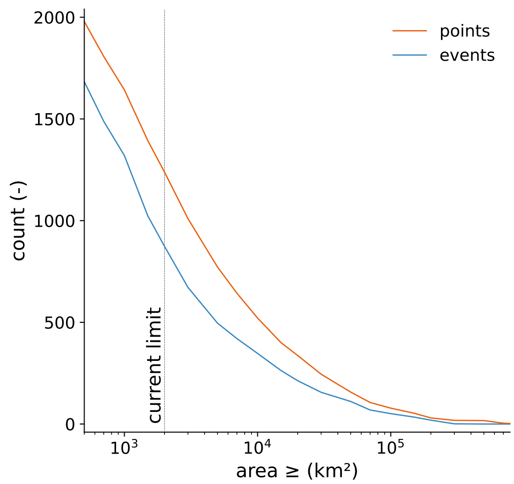

Figure 3: Number of stations (orange) and observed events (blue) by catchment area threshold. The vertical, dotted line represents a catchment area of 2000 km^2, the current limit in the EFAS notifications.

number of stations and "observed" events. Both counts decrease exponentially with catchment area. By excluding the points with contributing area smaller than 2000 km^2, we discarded 37% of the stations and 48% of the events. The objective of this decision was to use the current notification criteria as benchmark.

4.2. Individual NWP

Prior to studying how to combine the NWP models, we analysed the skill of each of the models individually. This analysis serves a dual purpose. First, to understand the relative skill of each model, with the intention of incorporating this skill into the computation of the total probability. Second, to compare the skill of individual NWP models against the benchmark—the current notification criteria—, and to set a baseline for the combination methods.

4.2.1. Probabilistic skill

In the initial assessment of NWP skill, we calculated the Brier score for each NWP at daily lead times. The Brier score, as an error metric, yields values that can be challenging to interpret, given the dependence of error magnitude on the specific variable—particularly in the context of rare events like floods, where the magnitude is inherently low. Instead, we employ the

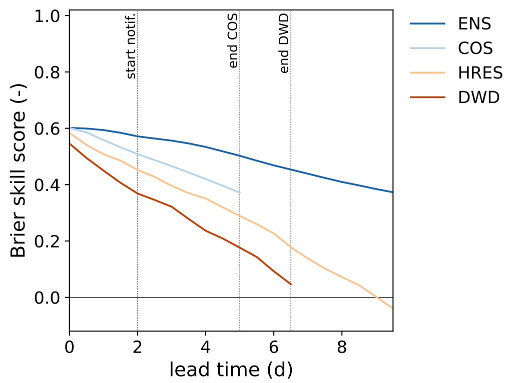

Figure 4: Brier skill score of the four NWP used in EFAS at daily lead times. The reference ($BSS = 0$) is a model that never predicts an event. Blue lines represent probabilistic models, and orange lines deterministic ones.

Brier skill score (BSS) to gauge the relative skill of a model in comparison to a reference. Models outperforming the reference receive a positive BSS, while those underperforming register a negative BSS. For the reference, we opted for a dummy model that never predicts an event ($P_{pred} = 0$) Legg and Mylne (2004).

The plot above demonstrates that probabilistic models exhibit greater skill than deterministic ones. Only within very short lead times (0-1 days) do deterministic models approach the skill observed in probabilistic models. This is particularly significant for EFAS notifications, as the formal notifications are reserved for lead times greater than 2 days (indicated by the leftmost dotted line). As lead time increases, there is a degradation in skill. This decline impacts ENS to a lesser extent than the other models. Both deterministic models display poor skill at their forecast horizon. For instance, at 10 days lead time, the skill of HRES is worse than a model that never predicts a flood.

Overall, the most skillful model for flood warning forecasting is ENS, while the least skillful is DWD. In section 4.3, we will convert the Brier score into weighing factors for the computation of total probability (see Figure 8).

4.2.2. Notification skill

In this section we analysed how the NWP would have performed if notifications would be sent based uniquely in one of them. First, we illustrate the general behaviour of notification skill according to the three dimensions here

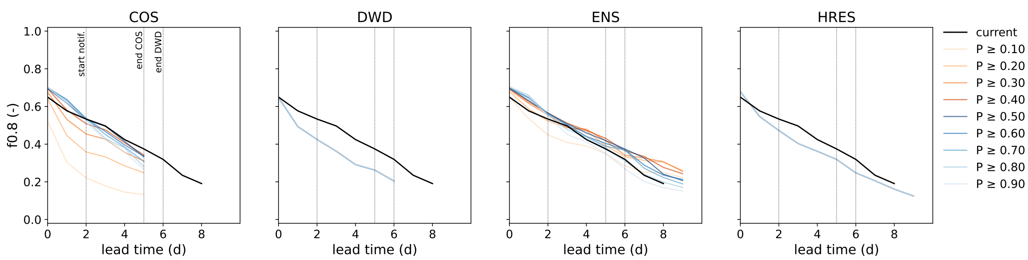

Figure 5: Evolution of the notification skill with daily lead times and probability threshold for each NWP model in an scenario with no persistence. Each plot represents a different NWP. As a benchmark, the black, solid line represents the skill of the current operational criteria (1 deterministic + 1 probabilistic, 30% probability and a persistence of 3 forecasts).

involved —lead time, probability threshold and persistence—. Second, we optimize the probability and persistence criteria for a fixed lead time range.

Figure 5 exhibits the evolution of the notification skill with lead time and probability threshold for a scenario with no persistence. The black, solid line is the notification skill of the current notification criteria, as a benchmark. Each colour line represents a different probability threshold; orange lines represent thresholds below 50% and blue lines over that value. For the two deterministic models (DWD, HRES) all lines overlap, since the probability threshold does not affect a deterministic forecast.

In line with Figure 4, the probabilistic models show higher skill than the deterministic models. Under certain conditions, they even outperform the current notification criteria, indicating potential for enhancing the skill of the current EFAS notification criteria. Deterministic models demonstrate skill on par with the current criteria solely for the initial lead time. As expected, notification skill degrades with lead time. This trend is consistent across all cases —deterministic, probabilistic and benchmark—, although the rate of degradation depends on the probability threshold. Lower probability thresholds sharply degrade at short lead times, while high probability thresholds experience a more pronounced loss of skill at longer lead times (such as ENS over 7 days).

There is a range of probability thresholds, approximately from 40% to 90%, exhibiting similarly good skill. Within this range, higher thresholds perform slightly better at short lead times, and lower thresholds perform slightly better at mid-long lead times. As we are seeking to identify a single optimal probability threshold for the notification criteria, the selection of the

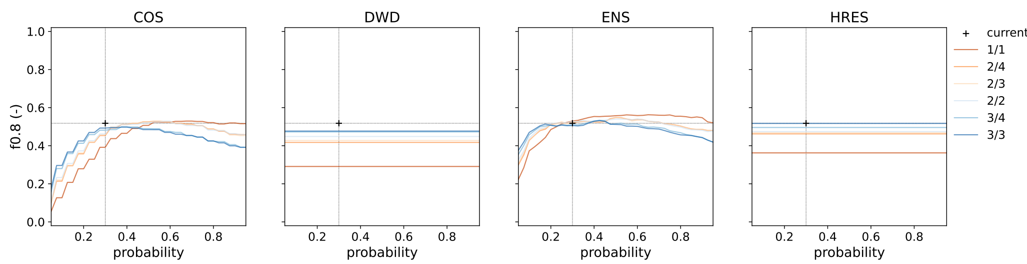

Figure 6: Evolution of the notification skill with probability threshold and persistence for each NWP model at a fixed lead time range (from 2 to 5.5 days). Each plot represents a different NWP model. As a benchmark, the black cross represents the current criteria (1 deterministic + 1 probabilistic, 30% probability and a persistence of 3 forecasts).

optimal criteria will prioritize a lead time range from 2 days (the minimum formal notification lead time) to 5.5 days (the maximum lead time in COS). This approach allows us to mitigate the substantial uncertainty associated with longer lead times, and focus on the lead times at which the majority of formal notifications are issued.

Up to this point, we have not yet considered persistence in our analysis. In order to assess the impact of persistence on the notification skill we have produced the plot in Figure 6, illustrating the evolution of skill with the probability threshold and persistence for a fixed lead time range (2-5.5 days). The benchmark (current notification criteria) is now represented by a single point corresponding the a 30% probability and a persistence of 3/3. As the probability threshold does not affect deterministic forecasts, the lines for the two deterministic models (DWD and HRES) are horizontal.

Three out of four NWP models are able to replicate or improve the skill of the current notification criteria at certain combinations of persistence and probability. Only DWD is less skillful than the benchmark. The skill of the probabilistic models shows a trade-off between probability threshold and persistence. Both criteria play the same role of removing false positives (or uncertain events) by taking stricter values. Consequently, stricter persistence necessitates lower probability thresholds to optimize skill, and vice versa. Nevertheless, the highest skill are consistently obtained in the scenario of no persistence. In this scenario, there is a wide range of probability thresholds (from 40 to 90%) that perform similarly well.

Persistence does have a significant impact in the deterministic models. Their skill improves dramatically as soon as some persistence is added in the

Table 3: Summary of the optimization of the notification criteria for individual NWP and four lead time ranges. The initial row serves as a benchmark, indicating the skill of the current notification criteria. In each lead time range, bold figures highlight the maximum obtained skill for a specific metric.

lead time	model	probability	persistence	recall	precision	$f_{0.8}$	rank
<2 d	1D+1P	0.300	3/3	0.522	**0.735**	0.634	3
	COS	0.875	1/1	0.673	0.718	**0.700**	1
	DWD	-	2/2	0.585	0.583	0.583	5
	ENS	0.800	1/1	**0.681**	0.711	0.699	2
	HRES	-	2/2	0.580	0.671	0.632	4
2-5.5 d	1D+1P	0.300	3/3	0.413	0.618	0.518	3
	COS	0.725	1/1	0.387	0.687	0.518	2
	DWD	-	3/3	**0.420**	0.522	0.477	5
	ENS	0.650	1/1	0.416	**0.725**	**0.562**	1
	HRES	-	3/3	0.416	0.612	0.517	4
5.5-7 d	1D+1P	0.300	3/3	0.215	**0.612**	0.356	2
	DWD	-	2/2	**0.291**	0.358	0.328	4
	ENS	0.450	1/1	0.264	0.605	**0.402**	1
	HRES	-	2/2	0.273	0.429	0.351	3
7-10 d	1D+1P	0.300	3/3	0.120	**0.625**	0.237	3
	ENS	0.175	2/2	**0.287**	0.396	**0.345**	1
	HRES	-	2/2	0.244	0.336	0.293	2

criteria. For the most stringent persistence (3/3), the skill of the deterministic HRES is as good as that of the probabilistic ENS, and even better than COS. However, as mentioned earlier, this is a sub-optimal value of persistence for the probabilistic models.

With the knowledge gained in the previous exploration, we were in a better position to understand the outcome of optimizing the notification criteria individually for each NWP, summarized in Table 1. The results are organised by lead time ranges; for each range, the table presents the optimized criteria —probability and persistence— and the skill metrics —recall, precision and f0.8— for the available NWP models. As a benchmark, the first row in each lead time range shows the skill of the current operational criteria.

The current EFAS setup (1D+1P) notifies 52% of the actual events (recall) and 74% of the notifications are correct (precision) at most. These figures reflect a negative bias, i.e., notifications are issued with a high level of certainty, leading to a limited number of false alarms, but many events being missed. This behaviour aligns with the selection of a biased target metric, such as the $f_{0.8}$ score, for the optimization of the notification criteria. The current criteria is not optimal as 1D+1P is consistently outperformed in terms of $f_{0.8}$ at any lead time by single NWP, suggesting that the combination of all NWPs hinders the skill ot that single NWP.

At the shortest lead time range, COS emerges as the top-performing NWP, closely followed by ENS. Both ENS and COS optimized a very high probability threshold and removed persistence. On the other hand, the skill of the two deterministic NWP is distinctively lower —particularly DWD—, and both require a persistence of 2/2. The current operational criteria is outperformed by any of the probabilistic models, and the deterministic HRES almost reaches that benchmark.

The scenario remains similar when examining lead times from 2 to 5.5 days. The two probabilistic NWPs continue to demonstrate higher skill than the deterministic ones, and outperform the current operational criteria. In this case ENS is the top-performing NWP. Neither of them require persistence and the probability thresholds are lower than in the previous lead time range, although still higher than the 30% value used currently. HRES, despite being deterministic, almost matches the skill of the benchmark and COS. As well as DWD, to maximize its skill requires a persistence of 3/3. Overall, there is a loss in skill of approximately 20% compared with the first 2 days of forecast.

At lead times from 5.5 to 7 days, the remaining probabilistic model, ENS, continues to outperform the other models. The optimal criteria follow the trend: no persistence and a decreasing probability threshold —though still higher than the benchmark—. Overall skill has degraded by 30% compared to the previous lead time range (over 40% compared to the first 2 days of forecast).

The skill at the longest lead time range is overall poor. Only two models have such a long forecast horizon, with ENS demonstrating significantly better skill. This long lead time range is the only one for which ENS requires persistence (2/2), a factor that notably decreases the probability threshold, reflecting the trade-off between these two criteria.

The Roebber diagram in Figure 7 offers an alternate perspective on the same results. Robber diagrams condense four skill metrics into a single plot Roebber (2009), with an ideal model positioned at the top right corner. Precision and recall are indicated on the X and Y axes, respectively. The slope of the dashed lines represents bias, calculated as the ratio of recall

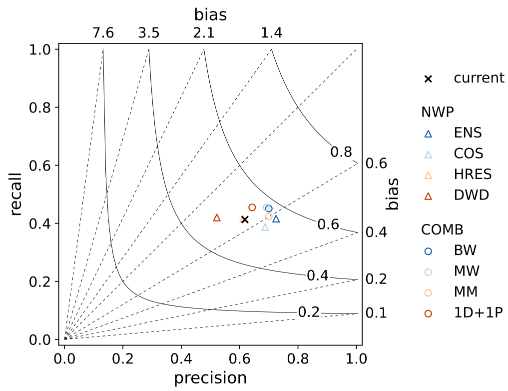

Figure 7: Roebber diagram of the optimized criteria at the lead time range from 2 to 5.5 days. Four skill metrics are represented: precision and recall in the X and Y axis, bias by the dashed lines, and the $f_{0.8}$ score by the solid lines. Triangles illustrate individual NWP and circles the combination of NWP. For reference, the skill of the current notification criteria is shown as a black cross.

to precision. Points below the 1:1 line indicate negative bias (a smaller number of notifications than observed events), while points above the line indicate positive bias (an excessive number of notifications). The $f_{0.8}$ score is represented by solid, black lines. Figure 7 shows simultaneously the skill of the current notification criteria, the optimization of individual NWP, and the optimization of combination methods (refer to Section 4.3) for the lead time range 2-5.5 days .

All NWP models and the current criteria exhibit negative bias, indicating that EFAS underpredicts the occurrence of flood events. In the case of the optimizations carried out in this analysis, the selection of the $f_{0.8}$ score as the target metric in the optimization forces this bias, as it prioritizes precision over recall. However, the current setup already shows a negative bias. It is remarkable that the improved $f_{0.8}$ of the probabilistic models is a result of higher precision, while recall values remain similar across all NWPs and the current criteria.

4.3. Combination of models

As explained in the methodology, three of the four combination methods compute total probability, which requires assigning weights to each NWP and lead time. Figure 8 illustrates the distribution of weights among NWP models for these three combination methods, and how the weights vary with

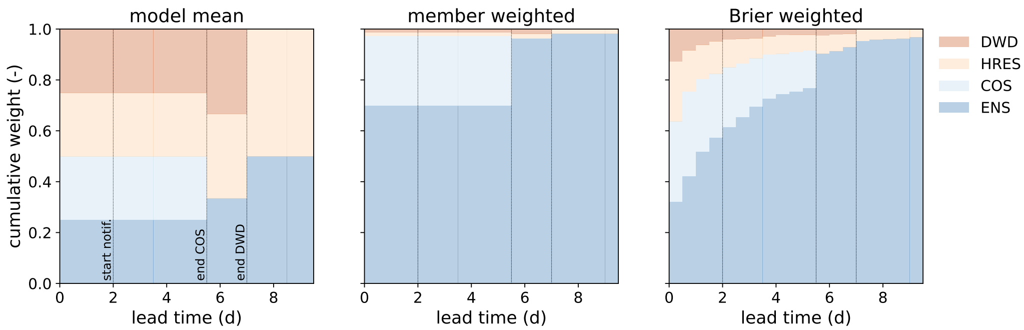

Figure 8: Weights applied to compute the total probability. Every plot represents a different methods, with colours representing NWPs (blue for probabilistic and orange for deterministic models).

lead time.

The model mean (MM) approach attributes equal weights to each model. The changes in the weights with lead time occur when the maximum lead time of a model is exceeded (5.5 days for COS and 7 days for DWD). This is the approach in which the deterministic models are given equal importance as the probabilistic models.

Instead, in the member weighted (MW) combination, every model run is assigned an equal weight. Therefore, ENS, which has a significant larger number of members, prevails over any other NWP. Even for the first 5 days, when all the models are available, ENS receives 70% of the weight. Conversely, the sum of the two deterministic models (HRES and DWD) has a maximum weight of 4% from days 5.5 to 7. In general, this approach heavily relies on the skill of the largest ensemble, which is not necessarily the most skillful, as is in the case of EFAS.

In the Brier weighted (BW) combination, we used the Brier scores from section 4.2.1 to derive weighing factors that prioritize the more skill models. The resulting weights vary with lead time, showing a transition from short to long lead times. At very short lead times, the deterministic models demonstrate skill, accounting for approximately a third of the total weight. However, as the lead time increases, the probabilistic models, particularly ENS, assume the majority of the weight. At lead times from 2 to 5.5 days, when most of the notifications are sent, the superior skill of probabilistic models is represented by a combined weight of approximately 80%.

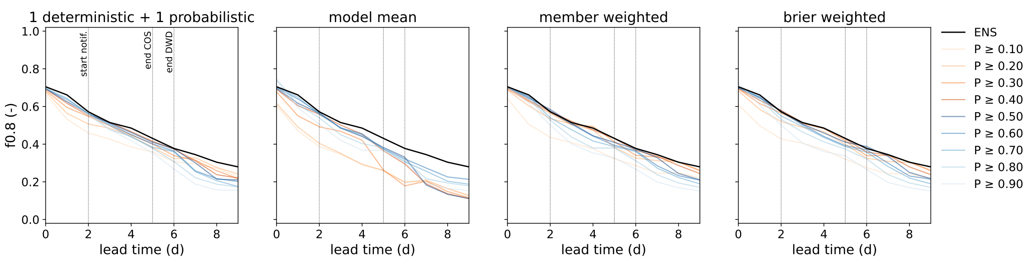

Figure 9: Evolution of the notification skill with daily lead times and probability threshold for each of the combination methods in an scenario with no persistence. Each plot represents a different combination of NWP. As a benchmark, the black, solid line represents the skill of ENS with optimized criteria.

4.3.1. Notification skill

Figure 9 and Figure 10 replicate the analysis of the notification skill previously done for the individual NWP models, but this time for the combination methods. Figure 9 illustrates the evolution of the notification skill with lead time and probability threshold for a fixed persistence value (no persistence). On the other hand, Figure 10 presents the evolution of skill with probability and persistence for a fixed lead time range (2-5.5 days). In both cases, the benchmark is ENS, the NWP that proved highest skill. Together these figures an overview of the behaviour of skill across four dimensions: combination method, lead time, probability threshold and persistence.

The results indicate that ENS is responsible for the majority of the skill in the grand ensembles, since only marginal gains are possible with specific combinations at certain lead times. Only two combination methods (MW and BW) are able to replicate the skill of the benchmark across all lead times, with minimal improvements at days 2, 4 and 6. Although MM at lead time 0 achieves the highest marginal gain, this analysis proves that this combination method is not suitable, as the skill of this grand ensemble is poorer than the benchmark from day 2 onwards. The current procedure (1D+1P) is also sub-optimal, even though the degradation with lead time is not as severe as with MM.

Regardless of the NWP or combination of NWP, the notification skill degrades notably with lead time. Even if at lead time 0 the skill is high (up to 0.74), the minimum lead time criterion limits the EFAS notification skill to values slightly lower than 0.6. The degradation continues and, from day 7 onward, the skill drops below 0.4, which may be an indicator for creating a

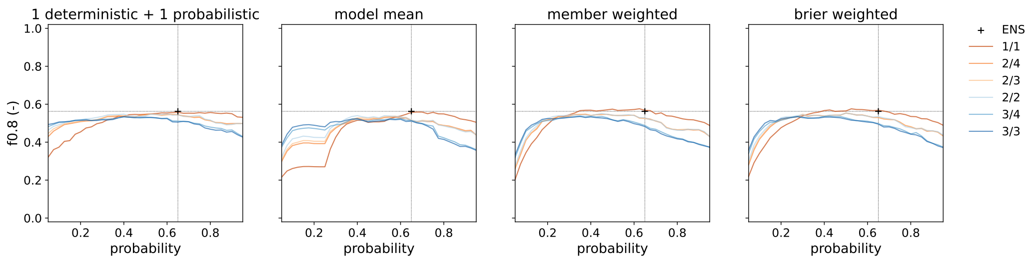

Figure 10: Evolution of the notification skill with probability threshold and persistence for each combination of NWP at a fixed lead time range (from 2 to 5.5 days). Each plot represents a different NWP model. Each plot represents a different combination of NWP. As a benchmark, the black cross represents the skill of ENS with optimized criteria (65% probability and no persistence).

new limitation on the maximum lead time at which notifications are issued.

Figure 10 reinforces the idea that the persistence criterion limits the maximum attainable skill in ensemble systems. Across all four combinations, the highest skill is reached when no persistence is applied (1/1). In this scenario, all combinations exhibit similar maximum skill. Only member weighted (MW) and Brier weighted (BW) marginally surpass the baseline (ENS). These two combinations show optimal skill within a range of probability thresholds spanning 40% to 70%. In both instances, the maximum skill is achieved at a probability threshold lower than that optimized for ENS.

The absence of sensitivity to probability in the current criteria (indicated by the dark blue line in the left-hand side pane) is noteworthy. Even at the lowest probability threshold, the $f_{0.8}$ score remains at 0.5, unlike other approaches where there is a significant decline in performance when the probability threshold takes very low values.

Table 4 summarizes the results of the optimization of the notification criteria for the combination methods in an identical way as Table 3 did for the NWP. The results are organised by lead time ranges; for each range, the table presents the optimized criteria (probability and persistence) and the skill metrics (recall, precision and $f_{0.8}$) for the four methods. As a baseline, the first row in each lead time range shows the skill of ENS. The objective of the optimization is again two-fold: to compare the four combinations of NWP models, and to assess the added value of building a grand ensemble instead of simply using the baseline.

In the shortest lead time range, two combinations clearly outperform the baseline (MM, BW). Between the two, MM exhibits the highest skill

Table 4: Summary of the optimization of the notification criteria for the combination methods and four lead time ranges. The initial row serves as a baseline, indicating the skill of ENS, the top-performing NWP. In each lead time range, bold figures highlight the peak value for a specific metric.

lead time	method	probability	persistence	recall	precision	$f_{0.8}$	rank
<2 d	ENS	0.800	1/1	**0.681**	0.711	0.699	4
	1D+1P	0.925	1/1	0.662	0.717	0.694	5
	MM	0.775	1/1	0.628	**0.838**	**0.741**	1
	MW	0.750	1/1	0.673	0.718	0.700	3
	BW	0.950	1/1	0.620	0.829	0.733	2
2-5.5 d	ENS	0.650	1/1	0.416	**0.725**	0.562	3
	1D+1P	0.600	1/1	**0.455**	0.643	0.554	5
	MM	0.700	1/1	0.424	0.700	0.559	4
	MW	0.500	1/1	**0.455**	0.692	0.574	2
	BW	0.525	1/1	0.451	0.700	**0.576**	1
5.5-7 d	ENS	0.450	1/1	0.264	**0.605**	**0.402**	1
	1D+1P	0.425	1/1	0.262	0.599	0.399	4
	MM	0.500	1/1	**0.276**	0.412	0.345	5
	MW	0.425	1/1	0.271	0.579	0.401	2
	BW	0.450	1/1	0.268	0.586	0.400	3
7-10 d	ENS	0.175	2/2	**0.287**	0.396	**0.345**	1
	1D+1P	0.250	1/1	0.256	0.414	0.334	4
	MM	0.225	2/2	0.252	0.334	0.296	5
	MW	0.175	2/2	0.278	0.402	0.343	3
	BW	0.300	1/1	0.255	**0.445**	**0.345**	1

($f_{0.8}$ = 0.741). This proficiency can be attributed to the importance that MM assigns to deterministic models, which demonstrated notable skill at very short lead times (see Figure 8). As of the optimized criteria, none of the four methods required persistence, and probability thresholds fell within the top quartile.

From day 2 to 5.5, BW and MW stand out as the top-performing combinations, with a comparable skill that surpass the baseline by 1%. Both approaches operate optimally without the need for persistence, with the probability threshold hovering around 50% (refer to Figure 10 for a sense of the uncertainty in defining the probability threshold). Compared with the preceding lead time range, there is an evident decline in skill by approximately 20%.

In the third lead time range, from 5.5 to 7 days, none of the combinations outperform the baseline, despite three of them (1D+1P, MW and BW) attaining very similar skill. None of these three methods necessitated the use of persistence, and the associated probability thresholds were slightly lower than those in the preceding lead time range (around 40%). When compared with the prior lead time range, there is a decrease in skill of around 30% (45% when compared to the the first range).

In the longest lead time range, BW emerges as the best-performing combination, with a skill on par with the baseline. MW, while slightly less proficient, shows similar skill. However, their optimized criteria differ. MW requires a persistence of 2/2 and a relatively low probability threshold (0.175) — precisely the same criteria optimized for the baseline. This results are not surprising, as at this lead time range MW and ENS are practically the same, since the weight of HRES in the grand ensemble is negligible. In contrast, BW does not require persistence, and its probability threshold (30%) aligns with the decreasing trend observed in previous lead time ranges.

The maps in Figure 11 illustrate the skill metrics associated with the optimal notification criteria, i.e., the BW combination with no persistence and a probability threshold of 52.5%. The metrics represent only the lead time range from 2 to 5.5 days, which corresponds to the period when the majority of the notifications are issued.

A pronounced contrast is evident in the maps, attributable to the limited number of events in most stations, resulting in skill metrics primarily yielding values of either 1 or 0 . In terms of f-score, there is a clustering of underperforming points in central Europe (notably the Alps) and central France. The breakdown into recall and precision reveals that the predominant cause

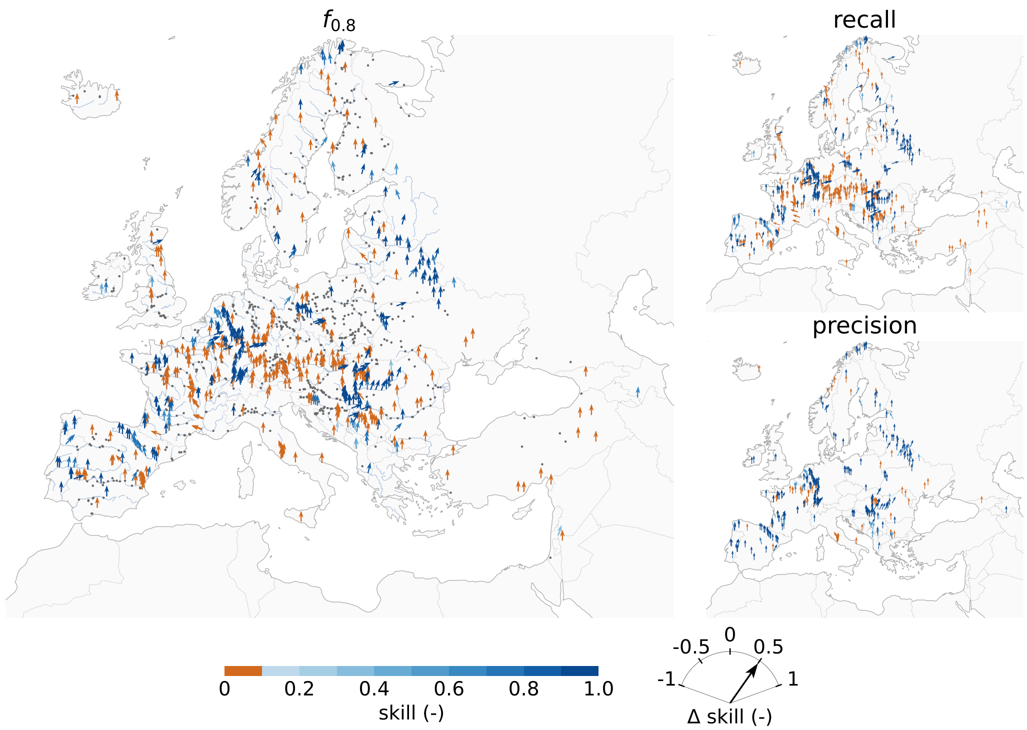

Figure 11: Skill maps for the optimized notification criteria (BW, 52.5% probability and no persistence) at the lead time range from 2 to 5.5 days. The main plot shows the skill in terms of $f_{0.8}$ (the target metric); in this plot, gray dots are stations for which the f-score cannot be computed due to all instances of hits, misses and false alarms being null. The smaller plots show precision and recall, the components of the f-score; for the sake of visibility, it only shows the stations for which the specific metric can be computed. In all the plots the direction of the arrows shows the difference in that metric between the optimized criteria and the current criteria; positive values imply gains and negative values losses in skill.

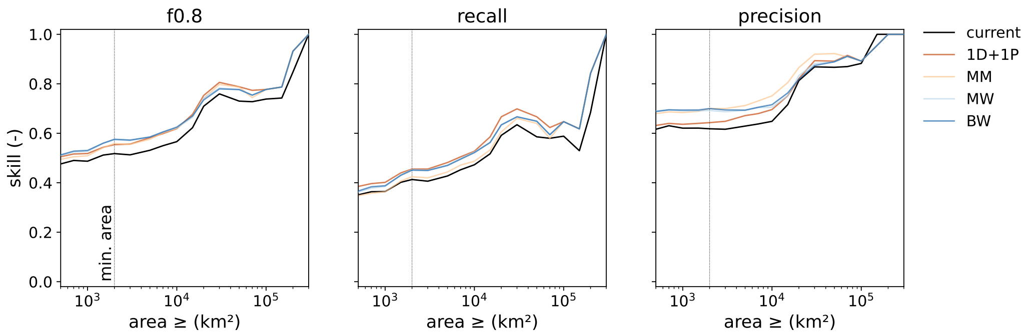

Figure 12: Evolution of skill with catchment area limit. Each plot represents a skill metric, with $f_{0.8}$ as the optimization target, and recall and precision as its components. In each of the plots, the black line represents the skill of the current notification criteria, while the colour lines depict the skill of the optimized criteria for the 4 combination methods. The vertical, dotted line indicates the catchment area limit currently applied in EFAS.

of the low f-score lies in recall, indicating missed events. Conversely, the precision of notifications remains consistently high, with exceptions observed in the Seine and Arno rivers. It is noteworthy the skill in all metrics for the Ebro, Rhine and Meuse—three rivers that experienced significant flood events during the study period.

By applying this set of criteria, EFAS would be able to issue notifications with at least 2 days lead time for 45% of the events, and 70% of the notifications would be correct. For reference, these figures represent an improvement in both metrics over the existing criteria, which achieved 41% recall and 62% precision (see Figure 7). The Roebber diagram indicates that the grand ensembles —namely MW and BW— still suffer from a negative bias, but it was slightly reduced by improving recall compared with ENS.

4.4. Skill by catchment area

The preceding sections presented results for a set of stations with a catchment area of at least 2000 km^2, the current limit. To investigate the potential relaxation of this criterion, Figure 12 illustrates the evolution of skill depending on the catchment area limit. The criteria used to create this figure are those optimized for the fixed area limit of 2000 km^2 and a lead time range between 2 and 5.5 days (refer to Table 4). For benchmarking purposes, the plots compare the skill of the four combination methods (coloured lines) with the skill of the current notification criteria (black line).

The skill of the system improves with catchment area, as expected. For very large catchments, all three metrics reach their maximum value of 1. However, the skill curves do not increase continuously. Recall experiences a loss in skill for areas ranging from 30,000 to 70,000 km^2, leading to a decrease in $f_{0.8}$ values. This loss is caused by missed events in the Guadiana, Seine, Loire, Rhone and Danube catchments. As already mentioned in previous sections, there is a notable gap between recall and precision, the components of the f-score, in both the optimized and current notification criteria, with precision consistently surpassing recall.

The vertical line at 2000 km^2 illustrates the gain in skill achieved through optimization, which is sustained across all catchment area values. Reducing the catchment area limit results in minimal loss in $f_{0.8}$, suggesting that this criterion could be reduced without compromising skill. For instance, the BW approach at 1,000 km^2 exhibits slightly higher skill ($f_{0.8} = 0.531$) than the current criteria at 2000 km^2 ($f_{0.8} = 0.518$). The relaxation of the catchment area limit impacts the f-score components differently. While precision remains unaffected, there is a loss in recall. To sum up, issuing flood alerts to smaller catchments would marginally reduce the skill of the system, primarily due to an increase in missed events, without leading to more false alarms.

In the previous analysis, a fixed probability threshold was applied for all catchments, regardless of their drainage area. We also investigated whether the skill of the system could be enhanced by adapting the probability threshold to the catchment area. Our rationale is rooted in the understanding that the level of uncertainty (model spread) differs between small and large catchments, suggesting that area-specific probability thresholds may enhance overall system skill. The results of this experiment are depicted in Figure 13. Each plot represents a different combination method, comparing the f-score (solid lines) and the probability threshold (dashed lines) between the model optimized for a 2,000 km^2 area limit (blue) and models optimized for increasing catchment area limits (orange). In order to compare the results of the two optimization runs, we kept a constant persistence value of 1/1, which was identified as optimal in Table 4.

The skill measured as $f_{0.8}$ is very similar for both optimizations, especially for areas smaller than 2,000 km^2. For larger areas, the main improvement is at the range between 30,000 and 100,000 km^2, for which the fixed criteria loses some skill. The evolution of the probability threshold with catchment area is erratic, likely due to the uncertainty in defining the optimal value (refer to Figure 10). Despite this, a general trend can be discerned, with

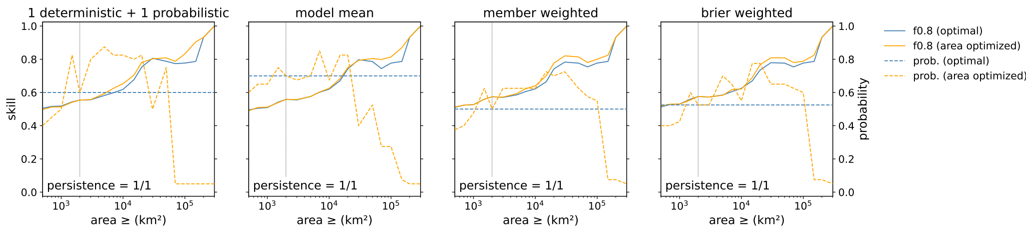

Figure 13: Comparison of the notification skill between a fixed (blue) and a area-specific (orange) probability threshold criterion for the lead time range from 2 to 5.5 days. Each plot depicts a different combination method, with solid lines representing the $f_{0.8}$ and dashed lines representing the probability threshold. The black, dotted, vertical line indicates the catchment area limit (2,000 km^2) used for optimizing the fixed probability threshold.

the threshold taking lower values for catchments smaller than 2,000 km^2, peaking at approximately 100,000 km^2, and dropping to a minimum of 5% for very large catchments.

Overall, the experiment indicates that varying the probability threshold does not significantly improve the skill of the notification criteria. Therefore, for the sake of simplicity, a fixed probability threshold is recommended.

5. Discussion

The definition of the spatial and temporal framework is often a limitation in studies that analyze the skill of ensemble flood forecasting systems Cloke and Pappenberger (2009). Due to the rarity of flood events, the statistical robustness of such analysis relies on a limited and often highly correlated sample of events. To address this limitation, long-term and continental or global studies are advocated, such as the one presented here, which spans over a period of two and a half years and extends over a European domain. Nevertheless, our study would benefit from a larger sample of points and flood events. As shown in Figure 11, there are many stations for which the skill metrics cannot be calculated due to null hits, misses or false alarms. A larger sample of observed events would reduce the susceptibility of the analysis to this issue. Two potential solutions emerge: extending the study period, which is limited by the availability of the HEPS forecasts (or reforecasts), or using an event threshold of higher recurrence. We decided to work with a sample of stations, instead of all the river cells Bartholmes et al. (2009), to limit the spatial correlation of events. Additionally, we tested removing stations with highly correlated discharge time series to further reduce correlation, but the results were similar and the sample of events much smaller; hence, this approach was discarded.

In the optimization of the notification criteria, we set a catchment area limit of 2,000 km^2, a threshold inherited from previous EFAS setups with lower spatial and temporal resolution. The results of this study (Figure 12) demonstrate that the current resolution of both NWP and hydrological models can produce skillful warnings at smaller catchments, which can significantly increase the number of stations and flood events analyzed, as shown in Figure 3. In line with this, the new EFAS version 5 (Joint Research Centre - European Commission (2023)) has increased spatial resolution from 5 km to 1 arc-minute, which is expected to further improve the skill of the system in smaller catchments. The skill of this new EFAS version will be analyzed as soon as a long enough period or reforecasts are available.

The selection of the target metric is a critical aspect in any skill analysis. Bartholmes et al. (2009) reviewed the desired characteristics of a skill score, and selected odds ratio, HK score and bias as their target metrics. In contrast, for our analysis, we selected the f_β score, a metric commonly used in machine learning, but rarely applied in hydrology. The f_β score has two interesting properties for analyzing a flood warning system. Firstly, it is based on the contingency table but avoids using true negatives, which could overestimate skill. Secondly, it allows for tailoring the notification criteria to the specific use case, as the β parameter can be adjusted to penalize stronger misses or false alarms. For instance, Bouttier and Marchal Bouttier and Marchal (2023) used the f_2 score to optimize the probability threshold for high-intensity precipitation warnings in France. The β value of 2 penalizes four times more missed events than false alarms, under the assumption that the cost of being hit by an extreme event unprepared is larger than the cost of mobilising resources in a false alarm. In our study, we took the opposite approach and decided to minimize the number of false alarms by using a β value of 0.8. This decision was based on the fact that EFAS complements the national or regional systems by issuing medium-range flood warnings to the responsible administrations, not to the general public. The aim is to alert authorities only when there is substantial evidence that the event will occur, in order to avoid excessive notifications that could undermine their trust in the system. Additionally, the event threshold (5-year return period) corresponds to a recurrence time shorter than the flood protection levels in many European regions, so we assume that missing minor events will not be as costly as Bouttier and Marchal argue. In a preliminary analysis, we tested β values of 0.8, 1 and 1.25 and found that the changes in the f-score were minimal (in the order of 3%) and in none of the cases persistence was needed. However, the selection of β affects the optimal probability threshold and the resulting bias, i.e., the distribution of errors between misses and false alarms. Values lower than 1 lead to higher probability thresholds and negative biases, whereas values larger than 1 yield positive bias, i.e., more notifications than actual events.

The first objective of this study was to evaluate the skill of the different NWP within EFAS, with a specific focus on comparing deterministic and probabilistic models. Figure 4 demonstrates that probabilistic NWPs outperform deterministic models in terms of Brier skill score across all lead times, with the exception of short lead times, where deterministic NWPs approach similar levels of skill. The outcome is similar when applying the notification criteria and measuring warning skill in terms of $f_{0.8}$ (Figure 5 and Figure 6). At their optimal notification criteria, the probabilistic NWPs, particularly ENS, outperform deterministic NWPs. The skill of both types of NWP degrades with increasing lead time, casting doubt on their ability to issue flood alerts more than 5-6 days in advance. The effects of the probability threshold and persistence criteria differs depending on the nature of the NWP. The probability threshold has a substantial effect on the skill of probabilistic models, while it does not impact deterministic models. Both COS and ENS perform better at probability thresholds over 50%, with a wide range of similarly performing values. On the contrary, persistence proves to be beneficial in improving the notification skill of deterministic NWPs, but it limits the skill of probabilistic models. Persistence, a.k.a. poor man's ensemble, is a method of creating an ensemble out of a deterministic forecast Cloke and Pappenberger (2009). For instance, a persistence of 3/3 is an ensemble of 3 members with a probability threshold of 100%. Our results show that while persistence removes the inherent erratic behaviour of deterministic models and notably improves their skill, it should not be applied to HEPS.

The second objective of this study was to determine the appropriate method to combine several NWP models into a grand ensemble. Careful consideration is necessary in selecting this method to avoid diminishing the skill of the most proficient NWP within the grand ensemble. For instance, the current EFAS notification criteria limit the skill of the system to that of the highest-performing deterministic model (HRES), which is poorer than any of the probabilistic NWP (Figure 6). This limitation is caused by a combination method (1D+1P) that excessively relies on deterministic forecasts, as well as the use of persistence. The challenge lies in finding a combination that enhances the skill of the best-performing NWP, which in the case of EFAS is the ENS.

Our findings demonstrate that ENS is a very strong baseline, with only a few combination methods yielding gains to its skill. The effectiveness of the grand ensemble is limited to the first 5 days lead time, when the total number of members is highest (73), particularly for lead times shorter than 2 days, when the deterministic models provide value. As an example, a simplistic approach like MM exhibits the highest skill within the shortest lead time range, potentially because it is the approach that assigns higher value to the deterministic models. Overall, the most promising approaches are the member-based (MW) and skill-based (BW) methods, as they outperform the baseline throughout the first 5 days lead time. Both methods have similar optimal notification criteria (no persistence and a probability threshold close to 50%), and their success is based on granting most of the weight to ENS. However, the reason why they allocate this weight to ENS differs. In the case of MW, it is coincidental that the most skillful model also possesses the largest number of members. On the contrary, BW grants greater value to ENS because it proved to be a superior model. If a more skillful model with fewer members would be added to the grand ensemble, BW would be able to extract that value, while MW would not. Therefore, we argue that skill-based combination methods are preferable.

The decision on the most appropriate combination methods can not be based solely on skill metrics, but also consider other factors such as the operational implementation effort, user comprehensibility, or the impact of adding or removing a NWP. Skill-based approaches, such as BW, are theoretically superior as the allocation of weights is based on quality rather than quantity. They are also the most robust when it comes to adding or removing NWP models. However, they are more challenging to implement and to explain to end users. To derive the matrix of weights (Figure 8), a skill-based approach requires having an extensive period of reanalysis and reforecasts upfront, posing a substantial computational burden on system implementation. Additionally, practitioners using the forecasts would require additional training to understand that the ensemble members are not equiprobable, which would be more intuitive, but there are some prevailing members.

There remains a question on the use of deterministic models in a grand ensemble dominated by a probabilistic NWP. When using a member-based combination, such as MW, deterministic models could be removed from the grand ensemble, as the weight allocated to them is so small that their inclusion barely affects the total probability. Skill-based approaches, instead, allocate larger weights to these models in the very first lead times, when they have proven value given their often higher resolution. The inclusion of deterministic models is a topic for future analysis, as the spatial resolution of deterministic and probabilistic NWP are nowadays similar, and centres such as the ECMWF plan to discontinue their deterministic models in favour of the ensembles.

6. Conclusions

This study describes a procedure to optimize warning thresholds in HEPS. We investigated the ability of deterministic and probabilistic NWP models in providing correct flood warnings, and we explored the effects of the notification criteria, namely probability threshold and persistence. The final objective of the study is to device a method to combine several NWP into a grand ensemble in order to outperform the most proficient, individual NWP. Our study case is the European Flood Awareness System (EFASv4), a medium-range continental flood warning system.

The individual analysis of the NWPs demonstrated that probabilistic models, particularly ENS, provide better flood warnings than deterministic models, even for short lead times, where the latter show some skill. We discovered that the persistence criterion is useful to increase the skill of deterministic models, as it creates a pseudo-ensemble, but is counterproductive in the case of ensembles. The criterion that maximizes ensemble skill is the probability threshold, whose optimum shows a certain degree of uncertainty, but always exceeds a value of 50%.

By default, the combination of several NWP into a grand ensemble does not improve the overall skill of the flood warning system. As an example, the current EFAS notification criteria, which uses four NWPs, is outperformed by only using one NWP (ENS) with the adequate criteria. Two of the approaches tested in this work yielded gains in skill compared with the baseline in medium and short lead times. One of these approaches (named member weighted) assumes equiprobability among all the model runs, and the other one (named Brier weighted) assigns weights to each NWP at every lead time based on their probabilistic skill. In our study case, these two approaches yielded similar results because the most proficient NWP is also the one with the largest number of members. Despite that, we argue that the skill-based approach is conceptually more appropriate, gives significance to the inclusion of deterministic NWP in the grand ensemble, and would perform better if the set of NWP would change. However, other study cases may consider other factors such implementation costs or explainability in the selection of the combination method.

Finally, we explored how catchment area affects the notification skill, with the idea of relaxing the current notification criterion that establishes a minimum catchment area. The results indicate that skill deteriorates with decreasing catchment area, as other studies report, but the improvements in meteorological and hydrological models over the last decade allow for a reduction in the minimum catchment area. We have proven that with the optimized criteria we can halve the area limit, therefore cover a larger area, and achieve a similar performance as in the current system setup. We expect that the skill at smaller catchments will be positively affected by the release of the new EFAS version with a higher spatial resolution.

7. Data availability

The reanalysis and forecast discharge data is available at the Climate Data Store of the Copernicus Climate Change Service. All the scripts, preprocessed data and results of the study can be found in this GitHub repository.

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